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S2**Started on** Thursday, 24 September 2026, 9:04 AM**State** Finished**Completed on** Thursday, 24 September 2026, 9:11 AM**Time taken** 7 mins 7 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  #include <string.h>
3
4  int max(int a, int b) {
5      return (a > b) ? a : b;
6  }
7
8  int main() {
9      char s1[1005];
10     char s2[1005];
11
12     if (scanf("%s", s1) != 1) return 0;
13     if (scanf("%s", s2) != 1) return 0;
14
15     int m = strlen(s1);
16     int n = strlen(s2);
17
18     int dp[m + 1][n + 1];
19     memset(dp, 0, sizeof(dp));
20
21     for(int i = 1; i <= m; i++) {
22         for (int j = 1; j <= n; j++) {
23             if (s1[i - 1] == s2[j - 1]) {
24                 dp[i][j] = 1 + dp[i - 1][j - 1];
25             } else {
26                 dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
27             }
28         }
29     }
30
31     printf("%d\n", dp[m][n]);
32
33     return 0;
34 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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